

JKBOSE 2026

CLASS 11TH

#XUV 

CHEMICAL

THERMODYNAMICS

LECTURE-2

CHEMISTRY

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ON
TOP



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Topics To Covered

- 1** *State and Path function.*
Types of Processes
- 2** Work

REVISION.

Thermodynamics

i) MOD for up N_2

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2px} = \pi_{2py} < \sigma_{2pz} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$$

ii) MOD for $Z > 7$ (O_2, F_2, Ne_2)

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2pz} < \pi_{2px} = \pi_{2py} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$$

BO =

$$\begin{array}{c} 16 \rightarrow 2 \\ 15 \rightarrow 2.5 \\ \boxed{14 \rightarrow 3} \\ 13 \rightarrow 2.5 \\ 12 \rightarrow 2 \end{array}$$

Magnetic property $\rightarrow$

Total $\bar{e}$ even = diamagnetic
(10 + 10) except
 $\bar{e}$ odd = paramagnetic

- $\rightarrow$ System.
- $\rightarrow$ Surrounding
- $\rightarrow$ Universe

Extensive + INTENSIVE PROPERTIES.

Chote-Chote topics

- System & Surroundings
 - Extensive & intensive
 - State & path function
 - Types of processes.
- 1, 2, 3 Marks

#STATE AND PATH FUNCTION

State of the System $\Rightarrow$ It simply means in which condition the system is present. The condition of the system is expressed in terms of pressure (P), Volume (V) & Temperature T .

• STATE FUNCTION $\Rightarrow$ Those properties of a system that depends upon only the initial and final state of the system not on the path followed by the system during change.

e.g. Pressure, Volume, Temperature, Internal energy, Gibbs free energy, entropy etc.

• PATH FUNCTION $\Rightarrow$

Those properties of a system that depends on the path followed by the system during change & not on the initial and final state of the system.

e.g. Work, Heat & Heat capacity.

QUESTION

Choose the correct answer. A thermodynamic state function is a quantity

- (i) Used to determine heat changes
- (ii) Whose value is independent of path
- (iii) Used to determine pressure volume work
- (iv) Whose value depends on temperature only.

STATE FUNCTION ✓

is

PATH
FUNK

QUESTION



For the process to occur under adiabatic conditions, the correct condition is:

(i) $\Delta T = 0$

(ii) $\Delta p = 0$

(iii) $q = 0$

(iv) $w = 0$

Types of Processes

(Important)

∂ = Small change
 Δ = large change



The state of a variable can be changed via thermodynamic process.

These processes are of the following types.

(a) Adiabatic Process: In adiabatic process, the system does not exchange heat with its surroundings i.e. no heat can flow in or out of the system.

For an adiabatic process, $dq = 0$ e.g. Sudden compression or expansion of gas.

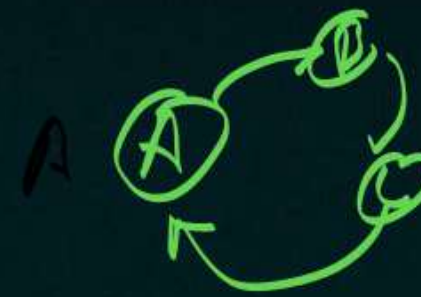
(b) Isothermal Process: The isothermal process is carried out at constant temperature i.e. it is the process in which temperature remains fixed.

For an isothermal process, $dT = 0$ e.g. slow expansion or compression of gas.

(c) Isobaric Process: In isobaric process, change of state is brought about at constant pressure i.e. $dp = 0$ e.g. Boiling of water.

(d) Isochoric Process: In an isochoric process, volumes of the system remains constant i.e. $dV = 0$ e.g. Heating of water in a closed container.

Types of Processes



(e) Cyclic Process: When a system is undergone a number of different changes and finally returns to its initial state, the process is called cyclic.

e.g. in a process shown below the system returns to the original state after successive changes.

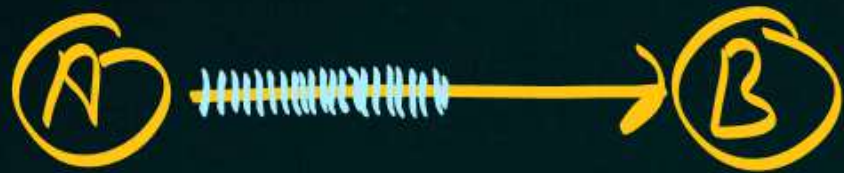


(f) Reversible Process: A reversible system is defined as a system where change takes place infinitesimally slow and the direction of which at any point can be reversed by small change in the state of the system. (P, V, T)



(g) Irreversible Process: An irreversible process is one, the direction of which cannot be reversed by small change in variables. All natural processes are generally irreversible and the net work done in an irreversible process is somewhat less than that in reversible process, i.e. $W_{irr} < W_{rev}$.

Rev



- Infinitely slow
- Infinite steps
- direction can be reversed easily by change (P, V, T).

Irr.



- Fast
- Finite step
- direction can't be reversed.

$$\underline{W_{rev}} > \underline{W_{irr}}$$

Work

(MECHANICAL WORK OR P-V WORK).

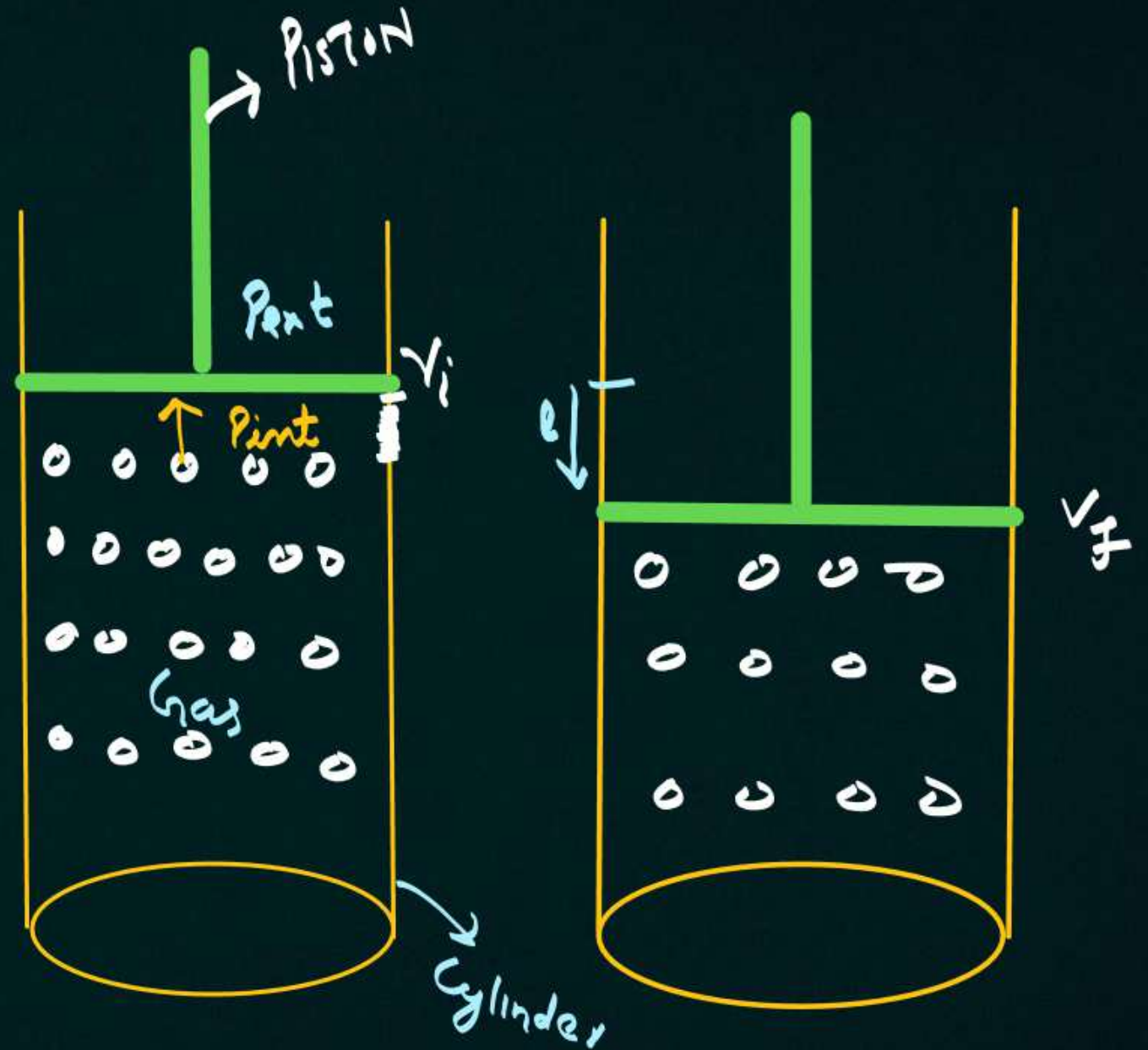
Let us consider a cylinder containing gas attached with a frictionless piston.

Now, we compress the gas from V_i to V_f , then some work is done.

This work can be calculated by two ways

a) Irreversible work done

b) Reversible work done.



Work



a) Irreversible work done.

When gas is compressed by applying external pressure (P_{ext}), the volume change can be written i.e. $\Delta V = l \times A$

$$\Delta V = V_f - V_i$$

Also we know that

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \text{ or } \text{Force} = \text{Pressure} \times \text{Area}.$$

$$\begin{aligned} \text{Work done} &= \text{Force} \times \text{displacement (l)} \\ &= \text{Pressure} \times \text{Area} \times \text{displacement} \end{aligned}$$

$$W_{irr} = P_{ext} \times A \times l$$

$$W_{irr} = P_{ext} \Delta V$$

From sign convention of the work.

$$W_{irr} = - P_{ext} \Delta V.$$

b) REVERSIBLE WORK.

As we know that in a reversible process there are infinite steps, therefore we have to calculate the work of each step by integrating.

$$\int dw = - \int_{V_i}^{V_f} P_{ext} dV$$

$$W_{rev} = - \int_{V_i}^{V_f} P_{ext} dV$$

For compression, $P_{ext} > P_{int}$ i.e. $P_{ext} = P_{int} + dp$

For expansion, $P_{ext} < P_{int}$ i.e. $P_{ext} = P_{int} - dp$

$$\text{i.e. } P_{ext} = P_{int} \pm dp$$

$$\text{i.e. } W_{rev} = - \int_{V_i}^{V_f} (P_{int} \pm dp) dV$$

$$\text{or } W_{rev} = - \int_{V_i}^{V_f} P_{int} dV \pm dp dV$$

$dp dV$ is the very small change thus can be neglected.

$$W_{rev} = - \int_{V_i}^{V_f} P_{int} dV$$

To be continued....

QUESTION



Summer Zone

[2026] ↙

Which of the following is a path function?

Path function

Work, Heat, Heat
Capacity

1 Internal Energy

2 Work

3 Entropy

4 Gibb's free energy

- ① WAHID KASOOL
- ② Mehran Ismal Bhat
- ③ Toiba Ashraf.

REVISE KARENG Chaka-Chak.

Thank
You